

Janhavi Chaurasia

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EDUCATION

Master of Science in Engineering, Robotics

[Aug'26 - May'28]

University of Pennsylvania, GRASP Lab

- Coursework: Machine Perception, World Models, Principles of Deep Learning

Bachelors of Technology in Computer Science Engineering

[Sep'22 - Apr'26]

School of Computer Science at Vellore Institute of Technology Chennai Campus

CGPA: 9.14/10 [3.901/4]

- Coursework: Cognitive Robotics, Machine Vision, Artificial Intelligence, Natural Language Processing, Linear Algebra, Differential Equations, Discrete Math and Graph Theory, Data Structures and Algorithms

SKILLS

Language and Framework: C/C++, Python, ROS/ROS2, OpenCV, Arduino, TensorFlow, Pytorch, OpenAI-GYM, MAVLink

Simulation and Modelling: Gazebo, MatLab, Rviz, IsaacLab/IsaacSim; **Version Control & Containerization:** Git, Docker

RESEARCH & PUBLICATIONS

Border Cell Representations for Spatial Navigation – GRASP Lab, University of Pennsylvania

[Sep'26 - Present]

- Investigating how border-cell-like spatial representations can emerge for embodied navigation, with a focus on formulating computational questions around learned spatial representations.
- Developing the initial research framework in collaboration with Prof. Pratik Chaudhari.

Emergent Representations in Dexterous Grasping – GRASP Lab, University of Pennsylvania

[Sep'26 - Present]

- Exploring whether learned models of hand-object interaction settle into a small set of grasp patterns, and how closely these line up with human-defined grasp taxonomies.
- Working with Prof. Jiayuan Mao to scope the experiments and the training constraints that shape this structure.

Neuromorphic-Geometric 6D Pose Estimation through Spiking Graph Neural Networks & SE3-Equivariant Learning

(SGSPose) Published at IEEE Access [[Paper](#)][[GitHub](#)]

- Developed SGSPose, a novel 6D pose estimation architecture integrating spiking neural networks for temporal spike encoding with SE(3)-equivariant graph neural networks enforcing geometric consistency for robust pose regression.
- Designed an elegant pipeline converting RGB images into temporal spiking features, followed by SE(3)-equivariant graph message passing with orthogonalized rotation outputs for precise, physically consistent 6D pose estimation.
- Utilized a composite loss combining position error (Smooth L1), geodesic rotation distance, and ADD-S metric with adaptive weighting, optimizing translation and rotation simultaneously.
- Achieved over 90% improvement in median translation error on 7Scenes compared to PoseNet, with translation errors as low as 0.019m and average < 0.06m; rotation errors range 8.3° - 21.3° with ADD-S accuracy up to 96.2%.
- Conducted as part of a research internship at Centre for Cyber Physical Systems (CCPS) at VIT Chennai.

Dopamine In The Mitigation Of Catastrophic Forgetting In Reinforcement Learning

Published at 2025 5th International Conference on Machine Learning and Intelligent Systems Engineering [[Paper](#)] [[GitHub](#)]

- Developed dopamine-inspired reinforcement learning methods integrating Elastic Weight Consolidation, neuromodulated temporal difference updates, and Hebbian learning to stabilize DQNs across sequential Atari tasks.
- Introduced dopamine-gated synaptic consolidation achieving near-zero forgetting (Backward Transfer = 0) and >99% reduction in weight changes compared to baseline DQN.
- Achieved positive backward transfer (+10) with neuromodulated learning rate, indicating reinforcement of prior knowledge during new task learning.

SwarmFormer: Transformer-Based Multi-Robot Formation Control With Risk-Aware Fusion

Published at 8th International Conference on Robotics, Control and Automation Engineering 2025 [[Paper](#)][[GitHub](#)] [[Dataset](#)]

- Developed SwarmFormer, a goal-conditioned transformer policy utilizing hierarchical cross-modal attention and Gaussian Mixture Model outputs for robust decentralized multi-robot formation control.
- Fused odometry, CVaR-augmented LiDAR, formation goals, and decentralized gossip communication, training on a Gazebo-ROS2 dataset spanning 5 formation types, 4 swarm sizes, varied obstacle densities, and communication topologies.
- Achieved a test Mean Absolute Error (MAE) of 0.0767, improving 21% over MLP and 28% over RNN baselines.
- Demonstrated strong probabilistic modeling with Negative Log-Likelihood (NLL) of -3.67, enabling uncertainty-aware control in complex environments.

Autonomous Exploration Of Unknown Indoor Environments Using Multi-UAV System

Published at 2025 5th International Conference on Computer, Control and Robotics [[Paper](#)] [[GazeboSim](#)] [[Pygame](#)]

- Developed a coordinated multi-UAV swarm system for autonomous indoor exploration using a master-slave architecture with dynamic task allocation and safety-aware coordination via ROS2.
- Integrated PX4 autopilot, Gazebo simulator, and ROS2 middleware for high-fidelity UAV simulation, sensor fusion,

and GPS-denied navigation using LiDAR-based SLAM and Visual-Inertial Odometry.

- Designed master and slave drone algorithms for intelligent floor assignment, adaptive swarm formation, recursive frontier exploration, obstacle detection, and battery-aware autonomous exploration and safe return.

EXPERIENCE

Teaching Assistant, University of Pennsylvania

[Aug'26 - Present]

- CIS 5150: Fundamentals of Linear Algebra & Optimization, Prof. Jean Gallier
- CIT 5920: Mathematical Foundations of Computer Science, Prof. Nan Zheng

Robotics Intern, Accio Robotics, Bengaluru

[Jan'26 - Jun'26]

Explainable robot diagnostics

- Conducted research on explainable robotic diagnostics, leading the development of NS-DART, a neuro-symbolic framework combining graph neural networks, temporal representation learning, physics-informed reasoning, and LLM-based explanations for autonomous robotic systems.
- Audited and extended production robotic software frameworks to improve system reliability, maintainability, and integration across existing robot platforms.

Robotics Intern, Accio Robotics, Bengaluru

[May'25 - Jul'25]

Bin Picking and Multi-Robot Coordination System Development

- Developed an industrial bin picking pipeline using YOLOv8 few-shot instance segmentation on RGBD data, chosen after evaluating Mask R-CNN and vision-language models to balance real-time performance with budget and compute limits.
- Built an interactive annotation tool with SAM2 integration, providing polygon and magnetic lasso editing, undo-redo, and YOLO export to accelerate high-quality dataset creation.
- Engineered projection homography pipelines with RealSense sensors for precise alignment and spatial consistency.

Robotics Intern, Clearspot.ai

[Aug'24 - Dec'24]

UAV Simulation and Perception System Development

- Processed and integrated image metadata for precise georeferencing and mapping within the software framework.
- Developed a UAV simulation environment using PX4 and ROS2, customizing flight dynamics and control algorithms to meet project specifications.
- Implemented computer vision techniques with OpenCV to analyze real-time drone camera feeds, enabling feature extraction and validation of perception-based navigation.

SELECTED PROJECTS

Quadruped Gait Imitation Learning with Sensorimotor Feedback

[Apr'25 - Aug'25]

Sensorimotor Feedback Control for Legged Robots using Imitation Learning and Predictive Modeling [\[GitHub\]](#)

- Designed and tuned a biologically inspired Central Pattern Generator (CPG) controller, collecting 2,075 high-fidelity demonstrations with explicit phase info for imitation learning of robust quadruped gaits.
- Developed behavioral cloning using multi-layer perceptrons on 18-dimensional inputs (16 joint angles + 2 phase features), achieving a forward model RMSE of 0.00185 radians ($\sim 0.11^\circ$), ensuring stable trotting gait reproduction.
- Engineered a deep forward predictive model with sensorimotor feedback control to detect gait anomalies and apply corrections in real-time, improving forward progress recovery by 399% under perturbations without loss of stability.

Multi-Agent Reinforcement Learning for Swarm Pursuit Coordination

[Nov'24 - Mar'25]

Pac-Man Ghosts as Multi-Agent Reinforcement Learning Swarm Pursuers [\[GitHub\]](#)

- Developed multi-agent RL framework implementing IQL, QMIX, and MADDPG within a custom Pac-Man simulation representing a multi-robot swarm pursuit problem.
- Applied a team-oriented reward function to encourage cooperative behavior, stabilize learning, and handle scalability and non-stationarity challenges.
- Achieved 100% and 90% win rates for QMIX and MADDPG respectively, with average episode lengths converging to ~ 48 steps; IQL lagged with 70% win rate and longer episodes (~ 73 steps).
- MADDPG delivered highest average rewards (~ 186) demonstrating most stable performance across 500 episodes.

Perception for Autonomous Underwater Vehicle

[Jan'24-May'24]

Perception system for autonomous underwater vehicles with ArUco pose estimation [\[Github\]](#)

- Developed a comprehensive underwater perception and navigation stack integrating ArUco-based pose estimation in ROS for accurate localization and mapping in dynamic environments.
- Laid the foundation for autonomous vehicle decision-making enabling precise navigation and task execution.

AWARDS & LEADERSHIP

- TAC Challenge Runners-Up & Team Leadership, Dreadnought Robotics, VIT Chennai

[Sep'23 - Aug'25]

As Head of Operations (2023-24), led the team to global Runners-Up at TAC Challenge 2024, Norway by overseeing coordination, technical and financial decisions, and developing ArUco marker-based perception and navigation stack for AUV. Advanced to Team Manager (2024-25), managing 120+ students, conducting component research and development methodology evaluation, overseeing finances and operations, and mentoring across robotics domains.